Lecture 16

Maxima and Minima of Multivariate Functions

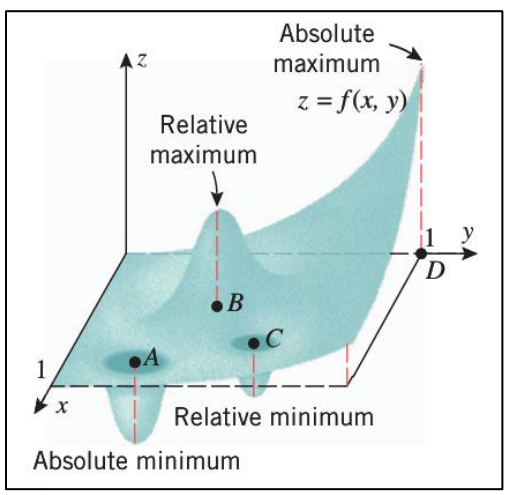
Relative Maximum: A function $f(x, y)$ of two variables is said to have a relative maximum at a point (x_0, y_0) if there is a disk centered at (x_0, y_0) such that $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) that lie inside the disk.

Relative Minimum: A function $f(x, y)$ of two variables is said to have a relative minimum at a point (x_0, y_0) if there is a disk centered at (x_0, y_0) such that $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) that lie inside the disk.

Absolute Maximum: A function $f(x, y)$ of two variables is said to have an absolute maximum at (x_0, y_0) if $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the domain of f.

Absolute Minimum: A function $f(x, y)$ of two variables is said to have an absolute minimum at (x_0, y_0) if $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) in the domain of f.

- If a function f has relative maximum or relative minimum at a point (x_0, y_0), then we can say that f has a relative extremum at (x_0, y_0).
- If a function f has an absolute maximum or absolute minimum at a point (x_0, y_0), then we can say that f has an absolute extremum at (x_0, y_0).



Critical Point: A point (x_0, y_0) in the domain of a function $f(x, y)$ is called a critical point of the function if $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$ or if one or both partial derivatives do not exist (undefined) at (x_0, y_0).

Saddle Point: A saddle point is a critical point which is not a relative extrema of the function.

Example: Let, $f(x, y) = y^2 - x^2$. Find the critical points.

Solution: Given, $f(x, y) = y^2 - x^2$.

So, $f_x = -2x, \quad f_y = 2y$

For critical points,

$f_x(x, y) = 0, \quad f_y(x, y) = 0$

$\Rightarrow -2x = 0, \quad \Rightarrow 2y = 0$

$\Rightarrow x = 0 \quad \Rightarrow y = 0$

Therefore, (0,0) is a critical point.

Since $f(x, y)$ is a polynomial, $f_x(x, y)$ and $f_y(x, y)$ exist everywhere. So, (0,0) is the only critical point.

<u>The Second Partials Test:</u>

<u>Theorem:</u> Let $f(x, y)$ be a function of two variables with continuous second-order partial derivatives in some disk centered at a critical point (x_0, y_0), and let,

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$$

i. If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a relative minimum at (x_0, y_0).

ii. If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a relative maximum at (x_0, y_0).

iii. If $D < 0$, then f has a saddle point at (x_0, y_0).

iv. If $D = 0$, then no conclusion can be drawn.

Example: Locate all relative extrema and saddle points of $f(x, y) = 4xy - x^4 - y^4$.

Solution: Given, $f(x, y) = 4xy - x^4 - y^4$

So, $f_x(x, y) = 4y - 4x^3$.

and $f_y(x, y) = 4x - 4y^3$.

For critical points:

$f_x(x, y) = 0 \qquad f_y(x, y) = 0$

$\Rightarrow 4y - 4x^3 = 0 \qquad \Rightarrow 4x - 4y^3 = 0$

$\Rightarrow y - x^3 = 0 \qquad \Rightarrow x - y^3 = 0$

$\Rightarrow y = x^3 \qquad \Rightarrow x = y^3$

So, we obtain,

$x = y^3$.

$\Rightarrow x = (x^3)^3$

$\Rightarrow x = x^9$

$\Rightarrow x^9 - x = 0$

$\Rightarrow x(x^8 - 1) = 0$

So, $x = 0 \quad \text{or} \quad x^8 - 1 = 0$.

Now,

$x^8 - 1 = 0$

$\Rightarrow (x^4)^2 - 1^2 = 0$

$\Rightarrow (x^4 + 1)(x^4 - 1) = 0$

$\Rightarrow (x^4 + 1)((x^2)^2 - 1^2) = 0$

$\Rightarrow (x^4 + 1)(x^2 + 1)(x^2 - 1) = 0$

$\Rightarrow (x + 1)(x - 1)(x^2 + 1)(x^4 + 1) = 0$

So,

$x + 1 = 0$	$x - 1 = 0$	$x^2 + 1 = 0$	$x^4 + 1 = 0$
$\Rightarrow x = -1$	$\Rightarrow x = 1$	$\Rightarrow x^2 = -1$	$\Rightarrow x^4 = -1$
		$\Rightarrow x = \pm\sqrt{-1}$	$\Rightarrow (x^2)^2 = -1$
		(complex value)	$\Rightarrow x^2 = \pm\sqrt{-1}$
			(complex value)

We only take real value for critical points. So, we get the following values of x,

$x = 0, \quad x = 1, \quad x = -1$.

$f(x, y) = 4xy - x^4 - y^4$ is a polynomial. So $f(x, y)$ is differentiable everywhere such that partial derivatives exist. So, we cannot find any value of (x, y) for which $f(x, y)$ is not differentiable. So, $f_x(x, y)$ and $f_y(x, y)$ exist for all values of x and y.

Now, we have, $y = x^3$.

if $x = 0,\ y = 0^3 = 0$

if $x = -1,\ y = (-1)^3 = -1$

if $x = 1,\ y = 1^3 = 1$.

So, (0,0), (1,1) and $(-1, -1)$ are critical points.

Now,

$f_{xx} = -12x^2$

$f_{yy} = -12y^2$

$f_{xy} = 4$

We get the following table:

Critical point (x_0, y_0)	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$	$D = f_{xx}f_{yy} - f_{xy}^2$
(0,0)	0	0	4	-16
(1,1)	-12	-12	4	128
$(-1,-1)$	-12	-12	4	128

At the points (1,1) and $(-1,-1)$ we have $D > 0$ and $f_{xx} < 0$. So, relative maxima occur at (1,1) and $(-1,-1)$. At (0,0), $D < 0$, so $f(x,y)$ has a saddle point at (0,0).

Maximum Value:

If $(x,y) = (1,1)$, $f(x,y) = 4 \cdot 1 \cdot 1 - 1^4 - 1^4 = 4 - 1 - 1 = 2$.

If $(x,y) = (-1,-1)$, $f(x,y) = 4(-1)(-1) - (-1)^4 - (-1)^4 = 4 - 1 - 1 = 2$.

Example: Locate all relative extrema and saddle points for $f(x,y) = xy + \frac{2}{x} + \frac{4}{y}$.

Solution: Given, $f(x,y) = xy + \frac{2}{x} + \frac{4}{y}$.

So, $f_x(x,y) = y - \frac{2}{x^2}$, $\quad f_y(x,y) = x - \frac{4}{y^2}$.

For critical points:

$$f_x(x,y) = 0 \Rightarrow y - \frac{2}{x^2} = 0 \Rightarrow y = \frac{2}{x^2}$$

$$f_y(x,y) = 0 \Rightarrow x - \frac{4}{y^2} = 0 \Rightarrow x = \frac{4}{y^2}$$

We obtain,

$$x = \frac{4}{y^2}$$

$$\Rightarrow x = \frac{4}{\left(\frac{2}{x^2}\right)^2}$$

$$\Rightarrow x = \frac{4}{\frac{4}{x^4}}$$

$$\Rightarrow x = x^4$$

$$\Rightarrow x^4 - x = 0$$

$$\Rightarrow x(x^3 - 1) = 0$$

So, $x = 0 \quad \text{or} \quad x^3 - 1 = 0$

Now,

$$x^3 - 1 = 0$$

$$\Rightarrow (x-1)(x^2 + x + 1) = 0$$

So,

$x = 1 \qquad \text{or} \qquad x^2 + x + 1 = 0$

Now,

$x^2 + x + 1 = 0$. So,

$$x = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$= \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$

For $x^2 + x + 1 = 0$, we find complex values of x. We ignore complex values and only take real values.

So, we get the following values of x,

$x = 0, \quad x = 1.$

If $x = 0, y = \frac{2}{x^2} = \frac{2}{0^2} =$ undefined.

If $x = 1, y = \frac{2}{x^2} = \frac{2}{1^2} = 2.$

Again, $f_x(x, y) = y - \frac{2}{x^2}, \qquad f_y(x, y) = x - \frac{4}{y^2}.$

Here, if $x = 0$, then $f_x(0, y) = y - \frac{2}{0^2} =$ undefined.

So, if $x = 0$, f_x does not exist. So, $(0, y)$ is a critical point

Here, if $y = 0$, then $f_y(x, 0) = x - \frac{4}{0^2} =$ undefined.

So, if $y = 0$ then f_y does not exist. So, $(x, 0)$ is a critical point.

So, $(1,2)$, $(0,y)$ and $(x,0)$ are critical points.

Now, $f_{xx} = \frac{4}{x^3}$, $\quad f_{yy} = \frac{8}{y^3}$, $\quad f_{xy} = 1$.

We get the following table:

Critical point (x_0, y_0)	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$	$D = f_{xx}f_{yy} - f_{xy}^2$
$(1,2)$	4	1	1	3
$(0,y)$	undefined	$\frac{8}{y^3}$	1	undefined
$(x,0)$	$\frac{4}{x^3}$	undefined	1	undefined

So, $(0,y)$ and $(x,0)$ are not valid critical points. At the point $(1,2)$, $D > 0$ and $f_{xx} > 0$, so $f(x,y)$ has a relative minimum at $(1,2)$.

Minimum Value:

If $(x,y) = (1,2)$, $f(x,y) = 1 \cdot 2 + \frac{2}{1} + \frac{4}{2} = 2 + 2 + 2 = 6$.

H.W. Locate all relative extrema and saddle points for the following functions:

i. $f(x,y) = x^2 + xy - 2y - 2x + 1$.

Answer: $(2,-2)$ is the only critical point. It is a saddle point.

ii. $f(x,y) = xy - x^3 - y^3$.

Answer: Critical points: $(0,0)$ and $\left(\frac{1}{6}, \frac{1}{12}\right)$. Saddle point at $(0,0)$. Relative maximum at $\left(\frac{1}{6}, \frac{1}{12}\right)$, with maximum value $= \frac{1}{432}$.

iii. $f(x,y) = x^3 + y^3 - 3xy + 4$.

Answer: Critical points: $(0,0)$ and $(1,1)$. Saddle point at $(0,0)$. Relative minimum at $(1,1)$, with minimum value $= 3$.

iv. $f(x,y) = 3x^2 - 2xy + y^2 - 8y$.

Answer: $(2,6)$ is the only critical point. It is a relative minimum, with minimum value $= -24$.

H.W. Exercise 13.8 (9–19)